

Question

Write a Java program to demonstrate the handling of `StringIndexOutOfBoundsException` using try-catch.

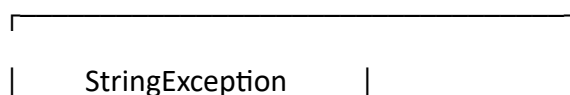
Steps for Execution

1. Create a package named `javapractice`.
2. Create a Java class named `StringException`.
3. Declare a String variable `str` and assign "Hello" to it.
4. Enter the try block.
5. Display "Accessing character from string".
6. Try to access the character at index 10 using `str.charAt(10)`.
7. The String "Hello" has only 5 characters, with valid indexes 0 to 4.
8. Therefore, Java throws `StringIndexOutOfBoundsException`.
9. The catch block catches the exception.
10. Display "String index is out of bounds".
11. Stop the program.

Algorithm

1. **Start**
2. Declare a String variable `str`.
3. Assign "Hello" to `str`.
4. Start the try block.
5. Access the character at index 10.
6. Since index 10 is outside the valid range, an exception occurs.
7. Catch `StringIndexOutOfBoundsException`.
8. Display "String index is out of bounds".
9. **Stop**

UML Class Diagram



|-----|

| + main(args: String[]): void |

|-----|

|

| handles



|-----|

| StringIndexOutOfBoundsException |

|-----|

| + StringIndexOutOfBoundsException |

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Code

```
package javapractice;
```

```
public class StringException {
```

```
    public static void main(String[] args) {
```

```
        String str = "Hello";
```

```
        try {
```

```
            System.out.println("Accessing character from string");
```

```
            System.out.println(str.charAt(10));
```

```
        }
```

```
        catch (StringIndexOutOfBoundsException e) {
```

```
            System.out.println("String index is out of bounds");
```

```
        }
```

```
    }
```

```
}
```

Output

Accessing character from string

String index is out of bounds

Result

The program successfully demonstrates `StringIndexOutOfBoundsException` handling using a try-catch block. The exception occurs because index 10 is outside the valid range of the string "Hello".